

Laser Thresholds and Symmetry-Breaking Pitchforks

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Notation: Laser variables and parameters

Let

$n(t) \geq 0$	number of photons
$N(t) \geq 0$	number of excited atoms
$G > 0$	gain coefficient
$k > 0$	photon-loss rate
$N_0 > 0$	excited-atom level without laser action
$\alpha > 0$	depletion coefficient

Idea: Photon gain and loss

Stimulated emission contributes GnN , while photon loss contributes $-kn$:

$$\dot{n} = GnN - kn.$$

Idea: Excited-atom depletion

Approximate the number of excited atoms by

$$N = N_0 - \alpha n,$$

so a larger photon population leaves fewer excited atoms available. Because $n \geq 0$ and $N \geq 0$, the physically admissible photon states satisfy

$$0 \leq n \leq \frac{N_0}{\alpha}.$$

Question: Where is the laser threshold?

Use $\dot{n} = GnN - kn$ and $N = N_0 - \alpha n$ to obtain a scalar equation. Find all equilibrium branches and classify the physically admissible states.

Solution. Substitution of $N = N_0 - \alpha n$ gives

$$\dot{n} = Gn(N_0 - \alpha n) - kn = (GN_0 - k)n - \alpha Gn^2.$$

Define

$$\mu = GN_0 - k.$$

Then

$$\dot{n} = f(n; \mu), \quad f(n; \mu) = n(\mu - \alpha Gn).$$

At the endpoints of the physical interval,

$$f(0; \mu) = 0, \quad f\left(\frac{N_0}{\alpha}; \mu\right) = -\frac{kN_0}{\alpha} < 0.$$

Uniqueness prevents crossing the equilibrium $n = 0$, and the vector field points inward at the upper endpoint. Thus $[0, N_0/\alpha]$ is forward invariant. The continuation criterion gives existence for all $t \geq 0$ because this interval is compact and the vector field is smooth.

The mathematical equilibrium branches are

$$n_0^* = 0, \quad n_1^* = \frac{\mu}{\alpha G}.$$

The physical parameters imply $\mu > -k$. Within this admissible parameter range, the branch n_1^* is defined for every μ , but it is a physically admissible photon state only when $\mu \geq 0$. Since

$$f_n(n; \mu) = \mu - 2\alpha Gn,$$

their rates are

$$f_n(0; \mu) = \mu, \quad f_n\left(\frac{\mu}{\alpha G}; \mu\right) = -\mu.$$

For $\mu = 0$, the equation on the physical state space is

$$\dot{n} = -\alpha Gn^2,$$

and its solution is

$$n(t) = \frac{n_0}{1 + \alpha Gn_0 t}.$$

Thus every $n_0 > 0$ converges to 0, while $0 \leq n(t) \leq n_0$ for every $t \geq 0$. Hence $n^* = 0$ is both attracting and Lyapunov stable at the nonhyperbolic threshold. On the full mathematical state space $n \in \mathbb{R}$, the two branches exchange stability at $\mu = 0$; the negative branch for $\mu < 0$ is not physically admissible. On the physical state space $0 \leq n \leq N_0/\alpha$,

$N_0 < k/G$	$n^* = 0$ is asymptotically stable
$N_0 = k/G$	$n^* = 0$ is nonhyperbolic and asymptotically stable
$N_0 > k/G$	$n^* = 0$ is unstable,
	$n^* = (GN_0 - k)/(\alpha G) > 0$ is asymptotically stable.

The pump threshold is

$$N_{0,c} = \frac{k}{G}.$$

Definition: Reflection symmetry

Let $I \subseteq \mathbb{R}$ be symmetric about 0, let $R \subseteq \mathbb{R}$ be a parameter set, and let $f : I \times R \rightarrow \mathbb{R}$. The scalar vector field has **reflection symmetry** if

$$f(-x; r) = -f(x; r) \quad \text{for every } x \in I \text{ and } r \in R.$$

Observation: Consequences of reflection symmetry

If $f(-x; r) = -f(x; r)$, then $x = 0$ is an equilibrium and every nonzero equilibrium occurs with its reflected partner. If f is five times continuously differentiable in x near $x = 0$, then, for each fixed r ,

$$f(x; r) = a_1(r)x + a_3(r)x^3 + a_5(r)x^5 + o(x^5) \quad (x \rightarrow 0),$$

where

$$a_1(r) = f_x(0; r), \quad a_3(r) = \frac{f_{xxx}(0; r)}{3!}, \quad a_5(r) = \frac{f_{xxxxx}(0; r)}{5!}.$$

Definition: Pitchfork bifurcation

A **pitchfork bifurcation** occurs when a persistent equilibrium branch changes stability while a reflected pair of nonzero equilibrium branches meets it.

Definition: Supercritical pitchfork

Orient the parameter so that the origin changes from asymptotically stable for $r < 0$ to unstable for $r > 0$. The pitchfork is **supercritical** if asymptotically stable nonzero branches exist for $r > 0$.

Definition: Subcritical pitchfork

With the same parameter orientation, the pitchfork is **subcritical** if unstable nonzero branches exist for $r < 0$.

Question: How does the supercritical normal form break symmetry?

Analyze

$$\dot{x} = rx - x^3 = x(r - x^2).$$

Solution. The origin is an equilibrium for every r . For $r > 0$, there are also two equilibria

$$x_{\pm}^* = \pm\sqrt{r}.$$

Since

$$f_x(x; r) = r - 3x^2,$$

we have

$$f_x(0; r) = r, \quad f_x(\pm\sqrt{r}; r) = -2r.$$

Therefore

r	$x^* = 0$	$x^* = \pm\sqrt{r}$
$r < 0$	asymptotically stable	absent
$r = 0$	asymptotically stable with algebraic decay	coincident at 0
$r > 0$	unstable	asymptotically stable

Question: How does the supercritical normal form break symmetry? (continued)

At $r = 0$, the solution of $\dot{x} = -x^3$ is

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}},$$

so convergence is algebraic.

Visualization: Supercritical potential landscape

For $\dot{x} = rx - x^3$, a potential is

$$V(x; r) = -\frac{r}{2}x^2 + \frac{1}{4}x^4.$$

r	critical points of V	landscape
$r < 0$	$x = 0$	one quadratic well
$r = 0$	$x = 0$	one quartic well
$r > 0$	$x = 0, \pm\sqrt{r}$	barrier at 0, wells at $\pm\sqrt{r}$

Question: Why does the hyperbolic-tangent model have a pitchfork?

For $\beta > 0$, analyze

$$\dot{x} = -x + \beta \tanh x,$$

and locate and classify its bifurcation.

Solution. The equation is reflection symmetric, and the origin is always an equilibrium. Since

$$\tanh x = x - \frac{x^3}{3} + O(x^5) \quad (x \rightarrow 0),$$

we have

$$\dot{x} = (\beta - 1)x - \frac{\beta}{3}x^3 + O(x^5) \quad (x \rightarrow 0).$$

The origin has rate $\beta - 1$. It is asymptotically stable for $0 < \beta < 1$, nonhyperbolic and asymptotically stable for $\beta = 1$, and unstable for $\beta > 1$.

For a nonzero equilibrium,

$$x^* = \beta \tanh x^*, \quad \beta = \frac{x^*}{\tanh x^*}.$$

For $x > 0$,

$$\frac{d}{dx} \left(\frac{x}{\tanh x} \right) = \frac{\sinh x \cosh x - x}{\sinh^2 x}.$$

The numerator vanishes at 0 and has derivative $2 \sinh^2 x > 0$ for $x > 0$. Thus $x/\tanh x$ increases strictly from 1 to ∞ . Hence no nonzero equilibria exist for $0 < \beta \leq 1$, while exactly one reflected pair exists for every $\beta > 1$. Along this branch,

$$f_x(x^*; \beta) = -1 + \frac{\beta}{\cosh^2 x^*} = -1 + \frac{x^*}{\sinh x^* \cosh x^*} < 0$$

for $x^* > 0$, by the same inequality $\sinh x^* \cosh x^* > x^*$. The reflected equilibrium has the same derivative. Thus both nonzero branches are asymptotically stable, and the bifurcation at $\beta = 1$ is supercritical.

Question: What does the local subcritical normal form predict?

Analyze

$$\dot{x} = rx + x^3.$$

Solution. The origin has rate r . Nonzero equilibria satisfy

$$x^2 = -r,$$

so they exist for $r < 0$ and are

$$x_{\pm}^* = \pm\sqrt{-r}.$$

At either nonzero equilibrium,

$$f_x(x^*; r) = r + 3(x^*)^2 = -2r > 0.$$

Thus the nonzero branches are unstable. The origin is asymptotically stable for $r < 0$ and unstable for $r > 0$. At $r = 0$,

$$\dot{x} = x^3 \begin{cases} < 0, & x < 0, \\ > 0, & x > 0, \end{cases}$$

so the origin is nonhyperbolic and unstable. For $r \geq 0$, every nonzero solution moves away from the origin; the local cubic model contains no large-amplitude saturation.**Question: How does a quintic term saturate a subcritical pitchfork?**

Analyze

$$\dot{x} = rx + x^3 - x^5 = x(r + x^2 - x^4).$$

Solution. For a nonzero equilibrium, set

$$y = x^2 > 0.$$

Then

$$y^2 - y - r = 0, \quad y_{\pm} = \frac{1 \pm \sqrt{1 + 4r}}{2}.$$

The two amplitude levels meet at

$$r = -\frac{1}{4}, \quad y = \frac{1}{2}, \quad x = \pm\frac{1}{\sqrt{2}}.$$

The equilibrium counts are

$$\begin{array}{l|l} r < -1/4 & x^* = 0 \\ r = -1/4 & x^* = 0, \pm 1/\sqrt{2} \\ -1/4 < r < 0 & x^* = 0, \pm\sqrt{y_-}, \pm\sqrt{y_+} \\ r = 0 & x^* = 0, \pm 1 \\ r > 0 & x^* = 0, \pm\sqrt{y_+}. \end{array}$$

At a nonzero equilibrium,

$$\begin{aligned} f_x(x; r) &= (r + x^2 - x^4) + x(2x - 4x^3) \\ &= 2x^2(1 - 2x^2) = 2y(1 - 2y). \end{aligned}$$

Question: How does a quintic term saturate a subcritical pitchfork? (continued)

For $-1/4 < r < 0$, $0 < y_- < 1/2$, so the y_- branches are unstable. For every $r > -1/4$, $y_+ > 1/2$, so the y_+ branches are asymptotically stable. The origin has rate r . At $r = -1/4$,

$$f(x) = -x \left(x^2 - \frac{1}{2} \right)^2,$$

so $x = \pm 1/\sqrt{2}$ are half-stable and Lyapunov unstable. At $r = 0$, the origin is unstable and $x = \pm 1$ are asymptotically stable. Therefore

$$-\frac{1}{4} < r < 0$$

has three asymptotically stable equilibria: the origin and two distinct large-amplitude equilibria, one positive and one negative. Suppose r is varied slowly enough that the state follows its current asymptotically stable equilibrium branch until that branch loses stability or ends. As r increases, the origin loses stability at $r = 0$. As r decreases, the large-amplitude branches end at $r = -1/4$.

Question: What bifurcation is encoded by each leading equation?

For $j = 1, 2$, let $R_j(x, \mu)$ be continuous in (x, μ) , odd in x , and continuously differentiable in x , with

$$R_j(x, \mu) = O(x^5), \quad \partial_x R_j(x, \mu) = O(x^4)$$

as $x \rightarrow 0$, uniformly for μ near 0. Classify

$$\dot{x} = \mu x - 4x^3 + R_1(x, \mu), \quad \dot{x} = -\mu x + 2x^3 + R_2(x, \mu).$$

Solution. For the first equation, division by $x \neq 0$ gives

$$0 = \mu - 4x^2 + O(x^4).$$

The right side has derivative $-8x + O(x^3) < 0$ for small $x > 0$. It starts at μ and is negative at a fixed small positive endpoint when μ is close to 0. Hence there is exactly one small positive root for $\mu > 0$ and none for $\mu \leq 0$. Reflection gives the negative root. All branch expansions below are as $\mu \downarrow 0$. The equilibrium equation gives

$$x^2 = \frac{\mu}{4} + O(\mu^2), \quad x_{\pm}^* = \pm \frac{\sqrt{\mu}}{2} + O(\mu^{3/2}),$$

for $\mu > 0$. The rates are $f_x(0; \mu) = \mu$ and $f_x(x_{\pm}^*; \mu) = -2\mu + O(\mu^2) < 0$ for sufficiently small $\mu > 0$. The pitchfork is supercritical.

For the second equation, division gives

$$0 = -\mu + 2x^2 + O(x^4).$$

The right side has derivative $4x + O(x^3) > 0$ for small $x > 0$. The same crossing argument gives a reflected pair exactly when $\mu > 0$. The equilibrium equation gives

$$x^2 = \frac{\mu}{2} + O(\mu^2), \quad x_{\pm}^* = \pm \sqrt{\frac{\mu}{2}} + O(\mu^{3/2}),$$

for $\mu > 0$. The rates are $f_x(0; \mu) = -\mu$ and $f_x(x_{\pm}^*; \mu) = 2\mu + O(\mu^2) > 0$ for sufficiently small $\mu > 0$. With the standard parameter $R = -\mu$, these unstable nonzero branches exist for $R < 0$, so the pitchfork is subcritical.